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Short communication

Time pressure leads to inhibitory control deficits in impulsive violent offenders

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Abstract

Impulsive violent criminal behavior is often ascribed to problems with behavioral control. Such behavior could be a consequence of stronger pre-potent responses or a failure of inhibitory control. A countermanding task which allows dissociation of these two processes was used to examine whether impulsive violent offenders were found to exhibit impaired inhibitory control. An experiment with a time restriction on responding was performed to effectively limit the ability of the subjects to strategically control their error rate. The results showed that the performance of the impulsive violent offenders was only impaired in the time pressure condition. These findings suggest that impulsive violent behavior may be linked to abnormal modulation of the frontal cortical areas, areas thought to be involved in response inhibition by negative feedback or emotional responses, rather than frontal dysfunction *per se*.

Keywords: Impulsive violent behavior; Inhibition; Countermanding task

1. Introduction

Inappropriate violent behavior poses a significant social problem. There is increasing interest in the specific neural mechanisms that underlie different patterns of violent behavior as well as the social conditions that contribute to such behavior. The end results of both premeditated and impulsive violence are socially costly. In the present study, we investigated the mechanisms underlying impulsive violent behavior which could be a consequence of either a greater tendency to generate pre-potent responses or a deficit in the inhibition of such responses. Impulsive violence is likely to be a consequence of different factors than premeditated violence. Premeditated violence is characterized by emotional dysfunction and lack of guilt as well as antisocial behavior and instrumental aggression [20]. Therefore, premeditated violence may be due to inability to form associations between emotional cues and actions (see [7]) or inability to experience some specific emotions and simulate them (see [13]). Social species typically inhibit aggressive attacks when a conspecific displays submission cues. In humans, this capacity is shown in the context of distress cues such as sad or fearful facial expressions. Several models have been suggested to explain the causes of inhibition of pre-potent behavior. For instance, Gray's [19] personality model proposes a behavioral activation system (BAS) and a behavioral inhibition system (BIS), dysfunction in either of which may lead to violent behavior [19]. Blair [5] proposed that a violence inhibition mechanism (VIM) could be triggered by distress cues and lead to withdrawal responses and termination of aggressive behavior in order to avoid the negative feedback associated with such distress cues produced by the victims [5]. These models propose that an underactive BIS is involved in inhibitory deficits and an overactive BAS

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is related to impulsivity, which would also show a response inhibition deficit. In other words, impulsive violent behavior may reflect a core deficit in inhibitory control or in elevated impulsivity.

Evidence from go/no-go tasks, which can provide an index of inhibitory control in the context of motor responses, suggests that aggressive adolescents have deficits in action inhibition [26]. Also, Chen et al. [11] used a go/no-go task to investigate the inhibitory processes of impulsive violent offenders, as compared to that of normal, non-offender, and subjects. Using electrophysiological indices, they found that the N2 component was associated with response inhibition and the amplitude of this component was lower in the impulsive violent group compared to that of the normal controls [11]. However, it was difficult to distinguish impulsive violent offenders from controls with behavioral measures in the go/no-go task because the duration of the covert inhibitory process could not be explicitly derived from the reaction time data.

The present study employed a countermanding paradigm in which a response to a 'go' signal has to be withheld if a 'stop' signal is subsequently presented. The go signal was a goal-directed cue trigger a manual keystroke response (so involving detection, response choice and preparation of response) and the stop signal was as an occasional and unpredictable warning cue related to response inhibition (involving detection of the stop signal and the inhibition of planned response). The associated race model [10,29] was employed in order to obtain measures of both inhibitory and response generation processes in impulsive violent offenders. The race model, which views performance as a race between the response signal and a stop command, has been shown to adequately describe empirical data [31], and simulation studies have demonstrated that the measures obtained with this task are reliable [3]. Deficits in performance on this task have previously been seen in Tourette's disorder [28], attention-deficit hyperactivity disorder and conduct disorder [2,40]. Logan et al. [30] found that highly impulsive subjects showed longer stop-signal reaction times on this task [30]. To our knowledge, the present study is the first one to use the task to test the inhibitory abilities of impulsive violent offenders.

Previous findings have demonstrated differences in electrophysiological measures but not behavioral performance in impulsive violent individuals of this type. The aim of the present study was to test whether the inhibition mechanism of impulsive violent offenders and normal subjects could be differentiated at a behavioral level. In this study, we also manipulated time pressure in the task. This was done to see whether restricting the ability to strategically alter the task failure rate (i.e., reduce negative feedback) by responding more slowly would differentially modulate performance in impulsive violent individuals. Recently, Ramautar et al. [37] found that go RTs were decreased concomitant with a reduction in the probability of a stop trials being presented, i.e.: the go RTs were faster in the block with low stop probability [37]. The pattern of result indicated that when the go trials were presented more frequently, subjects would elicit a stronger response tendency and faster go RTs were seen. In our experiments, only 25% of trials were stop trials, which is similar to Ramautar et al's low-probability condition (20%, in their study). It is therefore reasonable to infer that the go RT in this task is an index of pre-potent response tendency and stronger pre-potent response tendency is thus reflected by faster go RTs (see also [31]). Based on Gray's model, BAS is related to establishment of a response tendency to obtain rewards. Following their model, it seems reasonable to argue that response initiation is a function of the BAS and a faster go RT reflects a stronger BAS. This opinion was consistent with the observation of Logan et al. [30] where there is a trend showing that the mean go RTs in subjects with high impulsivity were faster than those with low impulsivity [30]. They considered the go signals as the momentum for the impulse and pre-potent response tendency, even though there was no specific motivational stimulus in the stop-signal task. In our study we took the go RTs and SSRTs as indicators of the strength of BAS and BIS, respectively. We analyzed the performance of impulsive violent offenders and matched controls in the countermanding paradigm in two different conditions: with and without time pressure. If we consider that impulsive violent offenders may be characterized by a higher BAS activity, the go RTs of violent offenders might be expected to be faster than the controls. In contrast, if we consider that impulsive violent offenders are characterized by a lower BIS activity, the SSRTs of violent offender might be expected to be longer than those of the controls. Furthermore, time pressure was also used to force the subjects to develop an impulsive response style characterized by faster reaction times to go stimuli and more unsuccessful stops. In such a situation, the subjects need to recruit more inhibitory control to overcome the stronger go response tendency. It may be difficult for impulsive violent offenders to inhibit their go responses in this condition.

2. Materials and methods

2.1. Participants

Nine volunteer male impulsive violent offenders took part in each experiment (mean age 35.1 years in Experiment 1; mean age 37.7 in Experiment 2). Nine controls matched for age, gender, education and IQ also took part in each experiment (mean age 32.1 years in Experiment 1; mean age 34.4 years in Experiment 2). There were no differences between the two groups in Raven's Progressive Matrices test in Experiment 1 (mean score 19 in impulsive violent group, mean score 18.2 in control group, $t_{0.05(8)} = 0.957$, $P = 0.367$) and Experiment 2 (mean score 18.2 in impulsive violent, mean score 17.3 in control group, $t_{0.05(8)} = 1.152$, $P = 0.283$). All participants had no history of psychiatric, neurological illness and no psychiatric treatment. The impulsive violent offenders had committed more than one homicide, and the homicides had occurred on separate occasions. They were repeat offenders, and were serving sentences of fifteen years to life in prison. Because they did not use violent behavior to achieve a specific desired goal such as obtaining the victim's possessions or to increase status within a group hierarchy, their offenses could not be categorized as premeditated homicide. They had caused the death of another person using their hands, knives or sticks. The subjects did not include murderers who committed robbery or rape. Furthermore, criminals who committed homicides while they were intoxicated or under the influence of drugs were also excluded. These relatively strict criteria were used to minimize the variability of the experimental groups, with the aim of increasing the likelihood of finding effects, should they exist, despite the reasonably low number of subjects tested. The control subjects were not impulsive violent offenders and were matched for age, gender and education. Five of nine control subjects counterfeited documents; one of them not only counterfeited documents but also illegally occupied others' properties; two of them counterfeited money; the other one sold and purchased stolen goods.

All subjects were recruited from Taipei Prison in Taiwan. All had normal or corrected-to-normal vision, gave written informed consent before participating in the study, and both experiments were approved by the ethics committee of Taipei Prison, Taiwan.

2.2. Design and procedure

The countermanding task consisted of go and stop trials. At the beginning of each trial, a central fixation dot appeared for 500 ms. Following offset of the central fixation dot, a black dot was presented to the left or right of the fixation at 9° of eccentricity on the horizontal meridian. Subjects were required to make a keypress response on a response box with the left index finger when the dot was presented on the left or with the right index finger when the dot was presented on the right. On 25% of the trials (stop trials), the central fixation dot reappeared as an instruction to withhold responses to the peripheral target. Stop signals were presented with a delay of 50, 90, 130, 170, 210, 250, 290 or 330 ms. A low proportion of stop signals (e.g., 25%) is typically employed to discourage use of unwanted strategies, such as delaying the go responses in order to get a higher success rate on stop trials.

In Experiment 2, the task was identical with the exception that a time restriction was used, and subjects were required to press a button within 600 ms and fast responses were encouraged. This was expected to further limit any strategies involving response delays to reduce error rates. In each experiment, subjects received 1 practice block followed by 5 experimental blocks. Each experimental block included 256 trials and lasted approximately 25 min; the occurrence and order of the eight stop-signal presentation conditions was randomized.

2.3. Apparatus

Testing took place in a sound attenuated consulting room in the prison with the conditions and equipment the same for both experiments. Stimuli were presented on a 17 in. CRT screen using video resolution of 800 × 600 pixels and a vertical refresh rate of 100 Hz. The subjects sat 60 cm in front of the screen which was at eye level.

2.4. Analysis

Individual mean reaction times for the correct trials were analyzed after removal of outliers (defined as responses more than two standard deviations from the mean) from the RT distribution on a subject-by-subject basis. The estimation of the internal response time to the stop signal (SSRT) was calculated using the distribution of go signal reaction times and the probability of responding given a stop-signal delay (the inhibition function) in accordance with the race model of Logan and Cowan [32]. The latency of the stop process can be estimated from the start time, which is under experimental control in the form of stop-signal delays, to the end of the stop process, which is inferred from the observed go RT distribution. In this study, SSRTs for each stop-signal time were estimated using the integral method (Fig. 1; [21,22]) then averaged to produce an overall estimate of SSRT. Although this method assumes that the SSRTs are constant across trials, violation of this assumption does not significantly alter the outcome of the analysis [21,32]. Notwithstanding this, Band et al. [3] found observed SSRTs changed with stop-signal delays. Consequently, we calculated the average SSRT for stop-signal delays where the proportion of trials in which subjects failed to inhibit responding lay between 0.15 and 0.85 [3].

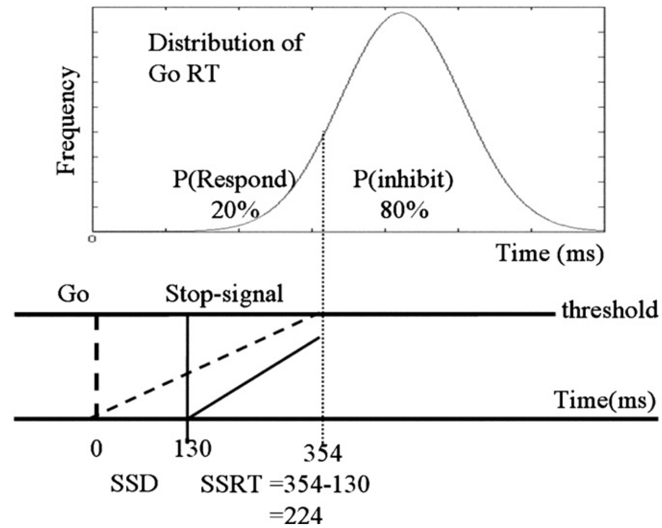

Fig. 1. Measuring stop-signal reaction time (SSRT). The distribution of go reaction times is integrated from the time of go signal presentation. For each stop-signal delay, a probability of responding is obtained. If the stop-signal delay of 130 ms resulted in a error rate = 0.20, meaning that the finish of the stop process should be equal to 20% of the go RT distribution. The point of 20% of the go RT distribution was 354 ms, and then the observed SSRT is 354 − 130 = 224 ms.

3. Results

3.1. The go reaction times and stop-signal reaction times

Fig. 2(A) and (B) shows the mean go RTs and SSRTs for the impulsive violent groups and the matched controls in Experiments 1 and 2. Analysis of go RTs used ANOVA with subject group and time limit as factors. A significant interaction effect was seen between the go RTs of the two groups across the different time pressure conditions ($F(1,32) = 5.833$, $P = 0.022$). The pattern of mean go RTs indicated that impulsive violent offenders showed faster RTs in Experiment 2 (with the time pressure) than in Experiment 1 (without time pressure) (394.2 ms vs. 517.2 ms, $F(1,8) = 13.876$, $P = 0.002$) whereas matched controls showed similar RTs in the two experiments (423.9 ms vs. 428.3 ms, $F(1,8) = 0.014$, $P > 0.05$). In Experiment 1 (without time pressure), the mean go RTs in

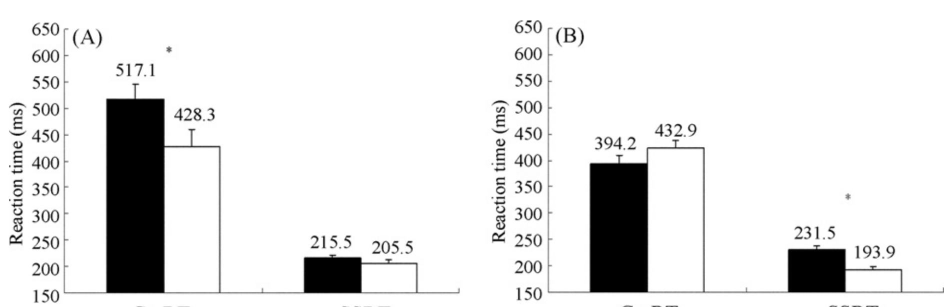

Fig. 2. Mean go reaction times, stop-signal reaction times and the inhibition functions. Panel (A) and (B) show the mean go reaction times and stop-signal reaction times for impulsive violent group (■) and control group (□) in Experiment 1 (A) and 2 (B) (*$P < 0.05$).

violent offenders (517.1 ± 29 ms) was longer than that of the matched controls (428.3 ± 32 ms), $F(1,8) = 4.08$, $P = 0.06$. In Experiment 2 (with time pressure), the mean go RT in violent offenders (394.2 ± 15 ms) was shorter than the matched controls (423.9 ± 15 ms), $F(1,8) = 1.858$, $P > 0.05$, but there was no significant difference between the two groups. The pattern of the results suggests that time restriction could prevent impulsive violent offenders from delaying their go responses since the impulsive violent offenders seemed to try to prolong their go response in order to perform better on stop trials in Experiment 1 but not in Experiment 2.

In the case of the SSRT variable, a significant interaction effect was seen between the SSRTs of the two groups across different time pressure conditions ($F(1,32) = 4.417$, $P = 0.044$). In Experiment 1, the mean SSRT also did not significantly differ between the violent offenders (215.5 ± 6 ms) and the matched controls (205.5 ± 8 ms), $F(1,8) = 0.925$, $P > 0.05$. However, the mean SSRT in the violent offenders (231.5 ± 6 ms) was significantly greater than that of the matched controls in Experiment 2 (193.9 ± 4 ms), $F(1,8) = 23.098$, $P < 0.05$. The pattern of mean SSRTs increased from the no time pressure condition to the time pressure condition in the impulsive violent offenders (215.5 and 231.5 ms, respectively), whereas mean SSRTs decreased from the no time pressure condition to the time pressure condition in the matched control group (205.5 and 193.9 ms). The results indicated that under time restriction, matched controls could inhibit their pre-potent responses more efficiently than impulsive violent offenders.

3.2. The inhibition function

A significant interaction was observed between the mean error rates of the two groups and different time pressure conditions ($F(1,32) = 9.445$, $P < 0.05$). As expected, error rates increased as the length of stop-signal delays increased. In Experiment 1, the mean error rate for violent offenders (0.26 ± 0.005) was lower than that of the matched controls (0.43 ± 0.008), but not significantly so ($F(1,8) = 3.408$, $P = 0.08$). In Experiment 2, the mean error rate in violent offenders (0.57 ± 0.005) was significantly higher than that of the controls (0.39 ± 0.005), $F(1,8) = 7.563$, $P < 0.05$. Impulsive violent offenders showed a higher error rate in the time pressure condition than in the no time pressure condition (0.57 vs. 0.26, $F(1,8) = 21.637$, $P < 0.05$) whereas the matched controls showed similar error rates in the two conditions (0.39 vs. 0.43, $F(1,8) = 0.199$, $P > 0.05$). Although the order of the tasks was fixed in both groups of subjects, no order effect was observed in the error rates and SSRTs. If an order effect occurred in this study, the error rates in Experiment 2 would be expected to be lower than those in Experiment 1 for both groups and the SSRTs in Experiment 2 should be shorter than those in Experiment 1. On the contrary, our results showed that the impulsive violent offenders showed a higher error rate in Experiment 2 (57%) than in Experiment 1 (26%), whereas the matched controls showed similar error rates in the two experiments. The absence of any change in performance in the controls suggests that the time pressure condition did not alter performance in these subjects or that time pressure counteracted the effects of repeated presentation. However, the pertinent comparisons are those involving the controls compared to the impulsive violent offenders. The impulsive violent offenders showed a longer SSRT in Experiment 2 (231.5 ms) than in Experiment 1 (215.5 ms) whereas the matched controls showed a shorter SSRT in Experiment 2 (193.9 ms) than in Experiment 1 (205.5 ms). Together, the results of these two experiments showed that the impulsive violent offenders made more errors on stop trials only when they were under time restriction conditions. The SSRTs in the impulsive violent offenders were significantly longer than matched controls especially in the time pressure condition (Experiment 2).

4. Discussion

Impulsive violent offenders are characterized by reckless behavior and are often incapable of controlling their behavior. The underlying causes of this lack of control could be that they make pre-potent responses too quickly or they are too slow to inhibit such behavior. We found that impulsive violent offenders and control subjects differed in performance on a countermanding task, which can be used to index these aspects of behavior. Critically, the nature of this difference depended on whether any time restriction was placed on responding. The pattern of results suggests that the BIS proposed in Gray's model [19] could be impaired in impulsive violent criminals.

In Experiment 1, slower responses were possible and utilizing a delayed response strategy would be expected to result in fewer errors on the task. The pattern of performance shows that impulsive violent offenders did make fewer errors under this condition than controls in addition to responding more slowly. Taken in isolation, these results suggest that impulsive violent offenders do not show a general tendency towards elevated prepotent responding (as would be reflected by reduced go RTs). The results from Experiment 2 show that when a time limit was placed on responding a different pattern of performance was seen; specifically the difference in go RTs was abolished. Elevated SSRTs were seen for the impulsive violent offenders and they made more errors. These results suggest that impulsive violent offenders do have problems with inhibitory control, but only under certain circumstances. A straightforward explanation of the pattern of results seen is that in Experiment 1 impulsive violent offenders adopt a strategy which minimizes errors and hence the degree of negative feedback. In Experiment 2 this strategy is no longer effective, as responding too slowly results in failure on a trial. Another interpretation would be that under conditions where momentum toward responding is greater, impulsive violent offenders have greater difficulty inhibiting responses. Experiment 2 might therefore be viewed as resulting in greater negative emotion/frustration due to either elevated error rates or because of the increased pressure involved in performance of the task. Frustration has long been linked to the display of reactive aggression [4]. This leads to reduced inhibitory control, which would reflect lesser engagement of the BIS in Gray's personality model. In the context of Blair's VIM, it may be that the same brain areas which are responsible for the altered performance seen on the countermanding task also underlie problems in inhibiting violent behavior. Thus, while the results do not allow determination of which of these models is more likely to account for the altered behavioral profile of impulsive violent offenders, they do offer insight into the possible underlying neural mechanisms.

Previous brain imaging studies [35,36] using positron emission tomography have found that impulsive murderers have lower left and right prefrontal functioning and lower right hemisphere prefrontal/subcortical ratios relative to the controls. In contrast, prefrontal functioning of predatory murderers was similar to that of the controls. Both impulsive murderers and predatory murderers were found to have higher right hemisphere subcortical activity than control subjects. Raine and colleagues suggested that excessive subcortical activity is a correlate of aggressive behavior, but predatory murderers have sufficiently good prefrontal functioning to regulate these aggressive impulses, while impulsive murderers were less able to regulate and control aggressive impulses due to deficient prefrontal regulation [36]. Consistent with this, Chen et al. [11] also found evidence of frontal cortex dysfunction reflected in the N2 amplitude at the Fz electrode in the impulsive violent group, with the N2 amplitude lower in offenders than for matched controls [11]. Thus, the elevated SSRTs found for the impulsive violent offenders may be related to frontal dysfunction.

The frontal lobe dysfunction hypothesis of violent behavior has been supported with studies of lesioned patients [1,6,18], neuropsychological and neuroimaging studies of individuals with antisocial behavior [8,34,35,36,41,43,46]. Earlier lesion and neuropsychological evidence consistently indicated that the reactive aggression was induced by the frontal lobe damage or dysfunction [6,18,34,35]. However, the precise subregions of the frontal lobe were not localized in these studies. Recently, some imaging studies of impulsivity-related personality disorders have related the poor inhibitory control of impulsive behavior to frontal dysfunction and also provided more precise underlying neural correlates of the impulsive behavior [14]. The converging evidence from these studies indicates that ventral and medial frontal cortex but not dorsalateral prefrontal cortex are correlated with the control of reactive aggression [9,12,18,42]. For example, Völlm et al. [44] found patients with impulsivity-related borderline personality disorder showed a more bilateral and extended pattern of activation across the medial, superior and inferior frontal gyri extending to the anterior cingulate cortex during response inhibition [44]. Patients with borderline personality disorder and antisocial personality disorder also exhibit a reduced regional cerebral blood flow in the superior frontal gyrus and the inferior frontal gyrus [17]. Horn et al. [23] used a go/no-go task and found subjects with low scores on Barratt's Impulsivity Scale showed greater activation in the anterior medial superior frontal gyrus [23]. These findings provide a better understanding of the frontal dysfunction that may result in impulsive violent behavior. To summarize, these studies have found that the inferior, orbitofrontal and superior medial frontal cortex were associated with the inhibitory control processes. We attempted to illustrate the relationship between the function of these brain areas and impulsive violent behavior by use of a task in which the involvement of these areas has been implicated. The inferior frontal cortex function has been linked to response inhibition (e.g. [16,24]), affective processing (e.g. [15]) and also monitoring the outcome of behavior [25,38,45]. Primate studies have also shown that lesions of the orbitofrontal cortex, but not medial prefrontal cortex, have disruptive effects on conditioned reinforcement [33,39]. However, Li et al. [27], using fMRI and minimizing confounds which might arise due to factors such as varying task demands and performance monitoring, found that greater response inhibition correlated with activation in superior medial and precentral frontal cortices [27]. This suggests that rather than having a deficit in functions associated with inferior frontal areas, which is a pathology associated with psychopathy [8], the impulsive violent offenders may produce violent behavior as a consequence of abnormal modulation of superior medial and precentral frontal cortices by affective information.

5. Conclusion

In summary, impulsive violent offenders do not show a general tendency towards elevated pre-potent responding. The SSRTs in the impulsive violent offenders were significantly longer than matched controls especially in the time pressure condition. Impulsive violent offenders showed deficits in inhibitory control under conditions where they were unable to strategically alter their failure rate. The neural basis of this deficit might be a consequence of abnormal modulation of the superior medial/precentral frontal cortex.

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